The Course Catalogues for the Courses offered in each basket are attached below:

Course Code: CSE 2007	Course Title: Data Structures and Algorithms Type of Course: Integrated	L- T- P- C	3 -0	2	4
Version No.	1.0				
Course Pre-requisites	Problem Solving Using Java				
Anti-requisites	NIL				
Course Description	This course introduces the fundamental concepts of data structures and to emphasize the importance of choosing an appropriate data structure and technique for program development. This course has theory and lab component which emphasizes on understanding the implementation and applications of data structures using Java programming language. With a good knowledge in the fundamental concepts of data structures and practical experience in implementing them, the student can be an effective designer, developer for new software applications.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Data Structures and Algorithms and attain Skill Development through Experiential Learning techniques.				
Course Out C omes	On successful completion of the course the students shall be able to: CO1: Implement program for given problems using fundamentals of data structures. [Application] CO2: Apply an appropriate linear data structure for a given scenarios. [Application] CO3: Apply an appropriate non-linear data structure for a given scenarios. [Application] CO4: Explain the performance analysis of given searching and sorting algorithms.				
Course Content:					
Module 1	Introduction to Data Structure and Linear Data Structure – Stacks and Queues	Assignment	Program activity		18 Sessions
Introduction – Introduction to Data Structures, Types and concept of Arrays. Stack - Concepts and representation, Stack operations, stack implementation using array and Applications of Stack. Queues - Representation of queue, Queue Operations, Queue implementation using array, Types of Queue and Applications of Queue.					

<table>
<tr><td>Module 2</td><td>Linear Data Structure-Linked List</td><td>Assignment</td><td>Program activity</td><td>17 Sessions</td></tr>
<tr><td colspan="5">Topics: Linked List - Singly Linked List, Operation on linear list using singly linked storage structures, Circular List, Applications of Linked list.

Recursion - Recursive Definition and Processes, Programming examples.</td></tr>
<tr><td>Module 3</td><td>Non-linear Data Structures - Trees and Graph</td><td>Assignment</td><td>Program activity</td><td>15 Sessions</td></tr>
<tr><td colspan="5">Topics: Trees - Introduction to Trees, Binary tree: Terminology and Properties, Use of Doubly Linked List, Binary tree traversals: Pre-Order traversal, In-Order traversal, Post - Order traversal. Graph - Basic Concept of Graph Theory and its Properties, Representation of Graphs.</td></tr>
<tr><td>Module 4</td><td>Searching & Sorting Performance Analysis</td><td>Assignment</td><td>Program activity</td><td>14sessions</td></tr>
<tr><td colspan="5">Topic: Sorting & Searching - Sequential and Binary Search, Sorting – Selection and Insertion sort.

Performance Analysis - Time and space analysis of algorithms – Average, best and worst case analysis.</td></tr>
<tr><td colspan="5">List of Laboratory Tasks:

Lab sheet -1

Level 1: Prompt the user, read input and print messages.Programs using class, methods and objects

Level 2: Programming Exercises on fundamental Data structure - Arrays based on Scenario.

Lab sheet -2

Level 1: Programming Exercises on Stack and its operations

Level 2: Programming Exercises on Stack and its operations with condition

Lab sheet -3

Level 1: Programming on Stack application infix to postfix Conversion

Level 2: -

Lab sheet -4

Level 1: Programming Exercises on Queues and its operations with conditions

Level 2: -

Lab sheet -5

Level 1: Programming Exercises on Linked list and its operations.

Level 2: Programming Exercises on Linked list and its operations with various positions

Lab sheet -6</td></tr>
</table>

Level 1: -   Level 2: Programming scenario based application using Linked List   Lab sheet -7   Level 1: Programming Exercises on factorial of a number   Level 2: Programming the tower of Hanoi using recursion   Lab sheet -8   Level 1: -   Level 2: Programming the tower of Hanoi using recursion   Lab sheet -9   Level 1: Programming Exercise on Doubly linked list and its operations   Level 2: -   Lab sheet -10   Level 1: Program to Construct Binary Search Tree and Graph   Level 2: Program to traverse the Binary Search Tree in three ways(in-order, pre-order and post-order) and implement BFS and DFS   Lab sheet -11   Level 1: Program to Implement the Linear Search & Binary Search   Level 2: Program to Estimate the Time complexity of Linear Search   Lab sheet -12   Level 1: Program to Implement and Estimate the Time complexity of Insertion Sort   Level 2: Program to Implement and Estimate the Time complexity of Insertion Sort   Lab sheet -13   Level 1: Program to Implement and Estimate the Time complexity of Selection Sort   Level 2: Program to Implement and Estimate the Time complexity of Selection Sort
Targeted Application & Tools that can be used   Use of PowerPoint software for lecture slides and use of Ubuntu for lab programs to execute. Tool is Codetantra tool.
Project work/Assignment:
Assignment: Students should complete the lab programs by end of each practical session and module wise assignments before the deadline.
Text Book   T1 Narasimha Karumanchi: "Data Structures and Algorithms Made Easy in Java", 5th Edition, CareerMonk Publications, 2017.

References

R1 Mark Allen Weiss: “Data Structures and Algorithm Analysis in Java”, 4th Edition, Pearson Educational Limited, 2014.

R2 Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser: “Data Structures and Algorithms in Java”, 6th Edition, John Wiley & Sons, Inc., ISBN: 978-1-118-77133-4, 2014.

R3 Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, 2017: “Introduction to Algorithms”, 3rd Edition, PHI Learning Private Limited.

Web resources:

For theory: https://onlinecourses.nptel.ac.in/noc20_cs85/preview

For Lab : codetantra tool

https://puniversity.informaticsglobal.com/login

Topics relevant to “SKILL DEVELOPMENT”: Llinked list and its type, Tree traversal and hashing tables for Skill Development through Experiential Learning techniques. This is attained through the assessment component mentioned in the course handout.